

Behind Your Serotonin

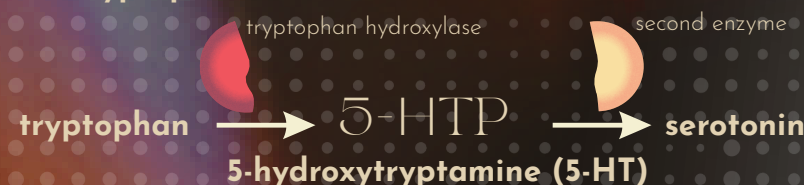


Serotonin is a neurotransmitter associated with mood, sleep, appetite, and other functions.

90% produced in the gastrointestinal (GI) tract, particularly by enterochromaffin cells in the gut lining.

10% synthesized in the brain, specifically in a group of neurons located in the Raphe nuclei, which are clustered in the brainstem

Serotonin, also known as 5-hydroxytryptamine (5-HT), is a monoamine neurotransmitter made from the essential amino acid **tryptophan**.



AMINO ACIDS are molecules that your body uses to make proteins. Tryptophan is one of the nine essential amino acids.

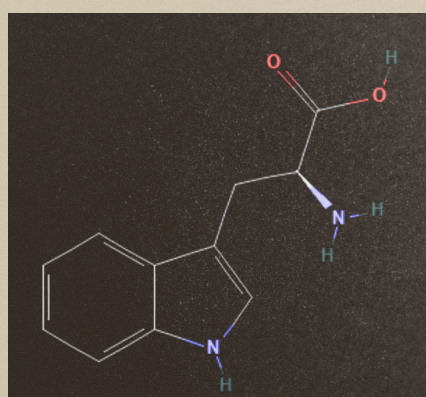
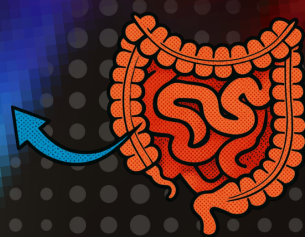
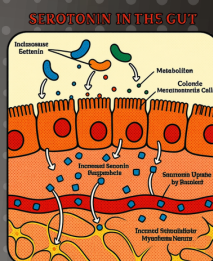
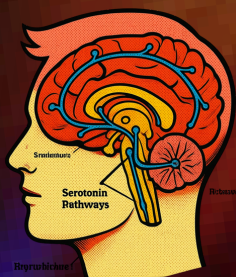
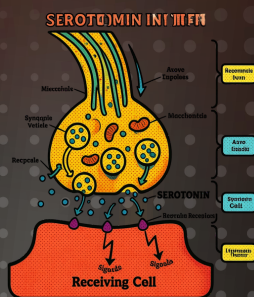
WHAT THEREFORE IS TRYPTOPHAN?

It has two isomers:

L-tryptophan is the form that your body uses to make proteins

D-tryptophan is the mirror image of L-tryptophan

but D-tryptophan doesn't work as well for the body as L-tryptophan



L-TRYPTOPHAN

IUPAC name: **(2S)-2-amino-3-(1H-indol-3-yl)propanoic acid**

Molecular Formula: **C₁₁H₁₂N₂O₂**

Molar Mass: **204.22 g/mol**

Melting Point:
554 to 558 °F (Decomposes) (NTP, 1992)
290.5 dec °C
282 °C (decomposes)
230 °C

L-tryptophan is a white powder with a flat taste. An essential amino acid; occurs in isomeric forms. (NTP, 1992)

White to slightly yellowish-white crystals or crystalline powder

Odorless

Slightly soluble in acetic acid, ethanol; insoluble in ethyl ether

Soluble 1 in 100 of water
very slightly soluble in alcohol
practically insoluble chloroform and ether
soluble in hot alcohol and solutions of dilute acids and alkali hydroxides.

Slightly bitter

Solubility in water:
0.23 g/L at 0 °C
11.4 g/L at 25 °C
17.1 g/L at 50 °C
27.95 g/L at 75 °C
49.9 g/L at 100 °C
13.4 mg/mL at 25 °C

Solubility
1 to 5 mg/mL at 68 °F (NTP, 1992)
13400 mg/L (at 25 °C)

USED ALSO IN:

Melatonin biosynthesis

Niacin biosynthesis



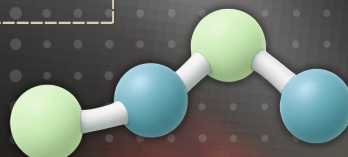
Functional Groups & Structural Feature

Amino group (-NH₂)

Carboxylic acid group (-COOH)

Indole ring containing a nitrogen atom

Chiral center at the α-carbon, giving L-tryptophan its specific stereochemistry



WHERE DO WE GET IT?



RELATION TO CHEMICAL ENGINEERING

Microbial fermentation

↓
Bioreactor
Control pH, temperature, oxygen, and mixing

↓
Separation & purification

↓
★ L-Tryptophan

L-Tryptophan can be produced through microbial fermentation. **Chemical engineers** help optimize the bioreactor conditions and downstream separation and purification processes to efficiently recover high-quality L-tryptophan from the fermentation broth.

RISK & SAFETY

- Not classified as hazardous by 222 of 225 ECHA notifications.
- May cause irritation upon contact with the eyes or skin.
- When heated to decomposition, it may release toxic fumes.
- It is reported as probably combustible, although flash-point data are unavailable.
- Teratogenic effects have been observed in experimental animals at high doses.
- Contaminated L-tryptophan supplements were linked to eosinophilia-myalgia syndrome (EMS).
- Avoid inhaling dust and wear gloves, safety glasses, and a lab coat when handling.

REFERENCES:

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